

Missing Data and Imputation

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Contents

Introduction	1
Load the required libraries.	1
Data Processing	2
Missingness Generation (SDD/MAR mechanism)	3
Inspection of missingness	4
Ad Hoc Imputation Methods	4
Listwise Deletion	4
Regression Imputation	5
Stochastic Regression Imputation	7
Multiple Imputation	8
Comparisson	11

Introduction

This markdown tackles ad-hoc imputation methods as well as multiple imputation, all using the `mice` package.

The following steps are performed:

- Data processing
- Inspection of the missingness pattern
- Apply ad hoc imputation methods using the `mice` package
- Apply multiple imputation using the `mice` package
- Perform linear regression and compare the output when different imputation approaches are used

Load the required libraries.

```
# Load libraries
library(tidyverse)
library(mice)
library(ggmice)
library(ISLR)

# set seed for reproducibility
```

```
set.seed(123)

# set scipen to suppress scientific notations
options(scipen = 999)
```

Data Processing

From the ISLR package, We will use the College dataset and the following variables:

- Out-of-state tuition
- PhD: Pct. of faculty with Ph.D.'s
- Terminal: Pct. of faculty with terminal degree
- Expend: Instructional expenditure per student

```
data_complete <-
  College |>
  select(Outstate, PhD, Terminal, Expend)

summary(data_complete)
```

```
##      Outstate      PhD      Terminal      Expend
##  Min.   : 2340   Min.   :  8.00   Min.   : 24.0   Min.   : 3186
##  1st Qu.: 7320   1st Qu.: 62.00   1st Qu.: 71.0   1st Qu.: 6751
##  Median : 9990   Median : 75.00   Median : 82.0   Median : 8377
##  Mean   :10441   Mean   : 72.66   Mean   : 79.7   Mean   : 9660
##  3rd Qu.:12925   3rd Qu.: 85.00   3rd Qu.: 92.0   3rd Qu.:10830
##  Max.   :21700   Max.   :103.00   Max.   :100.0   Max.   :56233
```

```
# This is the complete ground truth data before any missingness is introduced.
# It will be the benchmark against which every imputation method will be later
# compared to
```

We focus on how different imputation approaches affect the linear relationship between `Outstate` and `Expend`. First we inspect how the relationship looks like on the complete dataset.

```
# reg of Expend on Outstate using the full complete data
lm_complete <- lm(Expend ~ Outstate,
                  data = data_complete)

summary(lm_complete)
```

```
##
## Call:
## lm(formula = Expend ~ Outstate, data = data_complete)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
##  -6131  -2022   -637   1027  39273
##
## Coefficients:
```

```
##           Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 542.86969  385.92915   1.407        0.16
## Outstate    0.87325    0.03449  25.315 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3866 on 775 degrees of freedom
## Multiple R-squared:  0.4526, Adjusted R-squared:  0.4519
## F-statistic: 640.9 on 1 and 775 DF,  p-value: < 0.00000000000000022
```

```
# Outstate coefficient (0.873), highly significant p <0.0000000000000002
# This is the gold standard benchmark since no missingness exists yet
# This is the "correct" answer every later model will be compared against
```

Missingness Generation (SDD/MAR mechanism)

We can now generate missingness in the variable `Expend` and `Terminal` under a Seen Data Dependent (SDD) mechanism, assuming that the missingness in these variables are dependent on the values of the `Outstate` and `PhD` respectively

```
# logistic function
logistic <- function(x) (exp(x) / (1 + exp(x)))

# number of observations
N <- nrow(data_complete)

# predictor Outstate
misprob_out <- logistic(scale(data_complete$Outstate))

# predictor PhD
misprob_phd <- logistic(scale(data_complete$PhD))

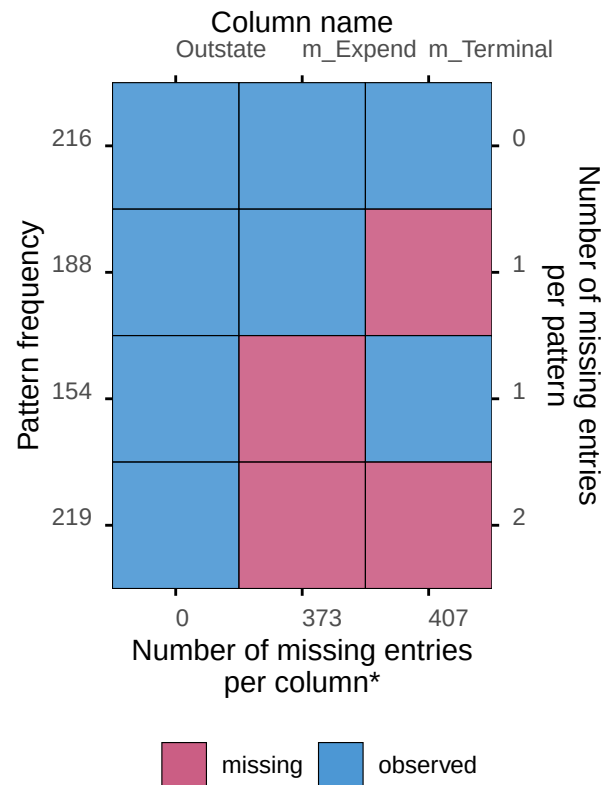
data_missing <-
  data_complete |>
  mutate(
    # response indicator outstate
    R_out = 1 - rbinom(N, 1, misprob_out),
    # response indicator PhD
    R_phd = 1 - rbinom(N, 1, misprob_phd),
    # Missing in expend
    m_Expend = ifelse(R_out == 0, NA, Expend),
    # and in terminal
    m_Terminal = ifelse(R_phd == 0, NA, Terminal)
  ) |>
  # Select variables that we will use
  select(Outstate, m_Expend, m_Terminal)

# Outstate (a variable that stays fully observed) is the driver of missingness
# in Expend, and PhD drives missingness in Terminal
```

Inspection of missingness

Inspect the missingness pattern using `ggmice` package function

```
# missingness pattern plot  
plot_pattern(data_missing)
```



*total number of missing entries: 780

```
# 219 rows are missing both variables (Expend and Terminal)  
# 154 rows are only missing Expend variable  
# 188 rows are only missing Terminal variable  
# 216 rows have complete data on all three variables  
# In total there are 780 missing entries across the dataset i.e  
# 373 missing in m_Expend (154 + 219)  
# 407 missing in m_Terminal (188 + 219)  
# Out of 1,554 possible missing cells for these two variables 2. (216 + 188 + 154 + 219)  
# meaning roughly half of all values in these two columns are missing.  
# Outstate itself has zero missing values (it's the fully-observed predictor of missingness).
```

Ad Hoc Imputation Methods

Listwise Deletion

This drops any row with a missing value and fits the regression on the 404 complete rows (777 - 373)

```
lm_listwise <- lm(m_Expend ~ Outstate,
                  data = data_missing)
```

```
summary(lm_listwise)
```

```
##
## Call:
## lm(formula = m_Expend ~ Outstate, data = data_missing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4986.2 -1490.9  -398.1   736.4 30053.4
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 2006.74472   396.63965    5.059 0.000000641 ***
## Outstate      0.70211     0.04153   16.904 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2758 on 402 degrees of freedom
## (373 observations deleted due to missingness)
## Multiple R-squared:  0.4155, Adjusted R-squared:  0.414
## F-statistic: 285.8 on 1 and 402 DF, p-value: < 0.00000000000000022
```

```
# Coefficient (0.702) vs true coefficient (0.873)
# The coefficient is a bit smaller,
# Standard error (0.0415) vs true standard error (0.0345)
# the standard error somewhat larger,
# so we introduced some bias, and reduced the power. Both could have been expected,
# due to how we generated missings and since we lost 373 cases.
# R squared (0.4155 vs true R squared (0.4526). It dropped.
```

Regression Imputation

This replaces each missing value with its predicted value from a regression model

```
imp_regression <- mice(
  data = data_missing,
  method = "norm.predict",
  m = 1,
  maxit = 1
)
```

```
##
## iter imp variable
## 1 1 m_Expend m_Terminal
```

```
# Coefficient (0.703). Almost identical to listwise deletion, still biased
# relative to the true coefficient (0.873)
# Standard error (0.0178), much smaller than the true standard error (0.0345)
```

```
# By imputing with no added noise, the method artificially shrinks the variability
# in the data, making standard error unrealistically small. In short, it is
# pretending to know these values more precisely than it actually does.
```

`complete()` takes an object, fills in the missing data, and returns the complete data in a specified format.

```
complete(imp_regression) |>
  is.na() |>
  sum()
```

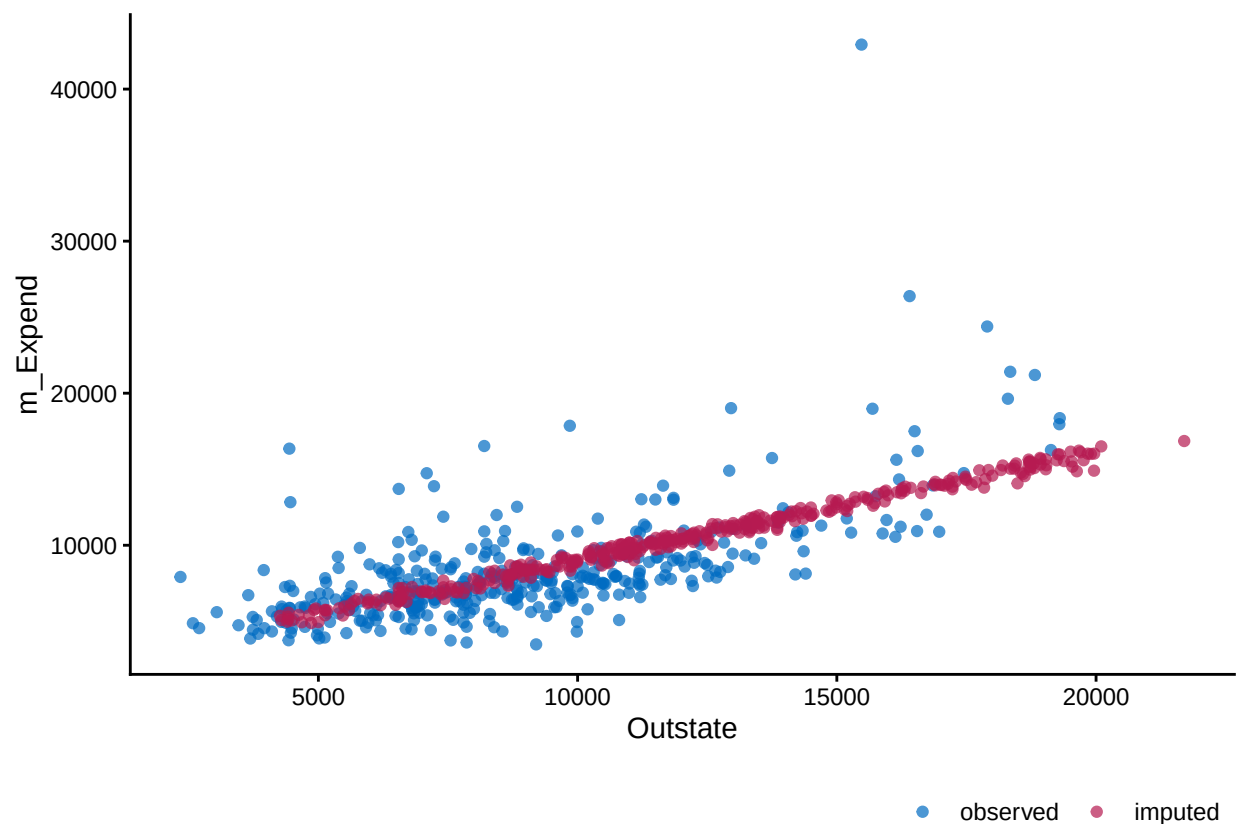
```
## [1] 0
```

```
lm_regression <- with(imp_regression, lm(m_Expnd ~ Outstate))
```

```
summary(lm_regression)
```

```
## # A tibble: 2 x 7
##   term      estimate std.error statistic    p.value  nobs df.residual
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl> <int>      <dbl>
## 1 (Intercept) 1989.      199.      9.97 4.05e- 22    777        775
## 2 Outstate    0.703     0.0178    39.5 1.44e-187    777        775
```

```
imp_regression |>
  ggmlc(aes(x = Outstate, y = m_Expnd)) +
  geom_point()
```



```
# Almost all imputed values are on a (almost) straight regression line (which
# is how we specified the imputation model). The variation around this line
# comes from the other variables that are also used in calculating the
# predictions.
# Each imputed value is a deterministic prediction with no residual variability
# added, all imputed points cluster tightly along the regression line rather
# than scattering naturally like real data would.
```

Stochastic Regression Imputation

This improves on plain regression imputation by adding random residual noise to each predicted value, so that imputed points scatter realistically around the regression line rather than sitting exactly on it.

```
imp_stochastic <- mice(
  data = data_missing,
  method = "norm.nob",
  m = 1,
  maxit = 1
)
```

```
##
## iter imp variable
## 1 1 m_Expend m_Terminal
```

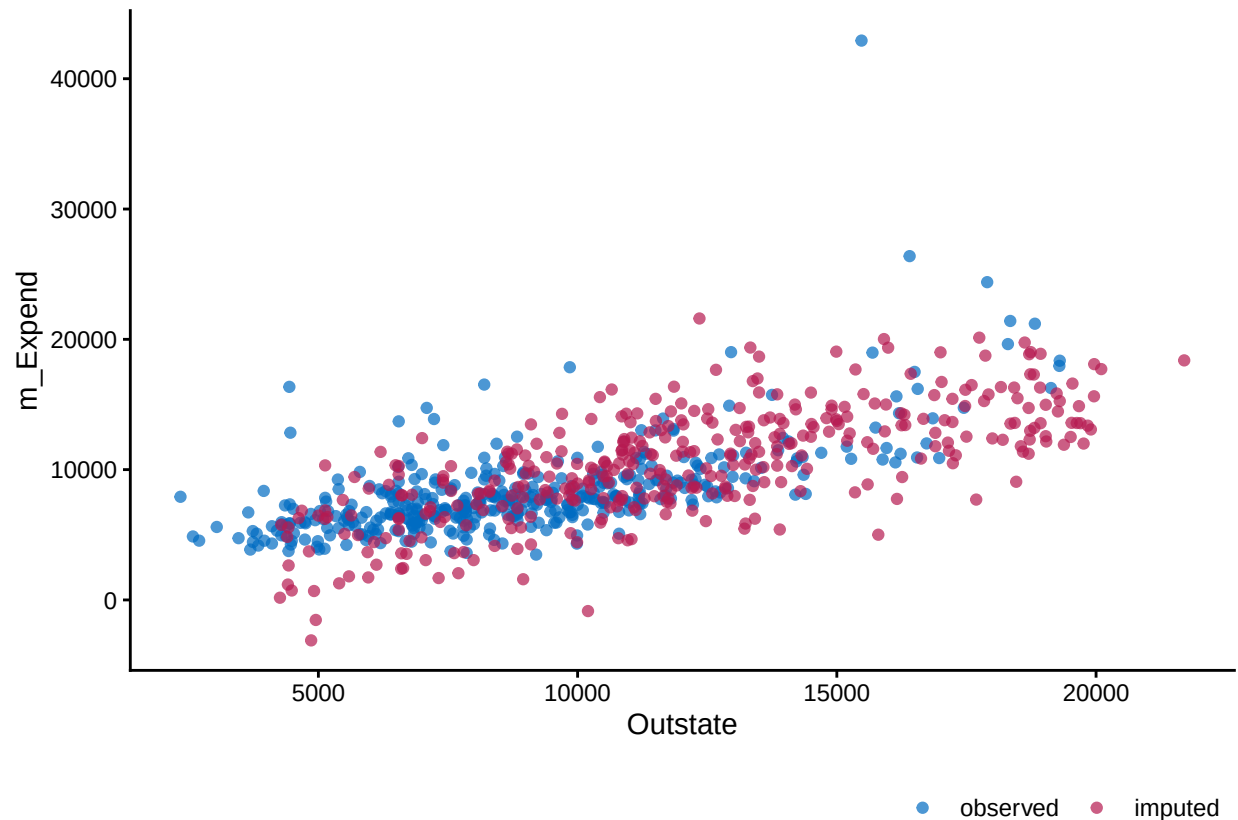
```
lm_stochastic <- with(imp_stochastic, lm(m_Expend ~ Outstate))

summary(lm_stochastic)
```

```
## # A tibble: 2 x 7
##   term          estimate std.error statistic  p.value  nobs df.residual
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl> <int>    <dbl>
## 1 (Intercept) 1873.      287.      6.53 1.22e- 10   777      775
## 2 Outstate    0.715     0.0257    27.9 6.66e-119   777      775
```

```
# Coefficients (0.715), similar to other ad hoc methods. Still biased.
# Standard error (0.0257), bigger than regression imputation (0.0178) but still
# smaller than the true (0.0345)
```

```
imp_stochastic |>
  ggmlm(aes(x = Outstate, y = m_Expend)) +
  geom_point()
```



*# There is considerably more variation between the imputed values as compared to
the regression imputations. However, the current approach still treats the
estimated regression coefficients as fixed, although there is also uncertainty
with respect to these coefficients. This uncertainty is still not taken into
account with stochastic regression imputation.*

Multiple Imputation

This creates 5 separate imputed datasets (the `m=5` default), each filled in using Predictive Mean Matching (“pmm”), iterating 5 times (`maxit=5`, the default) per dataset.

- 5 imputations, method “pmm” for both `m_Expend` and `m_Terminal`
- Predictor matrix: shows which variables are used to impute which, for example
Outstate and `m_Terminal` are both used as predictors when imputing `m_Expend`, and vice versa

```
imp_multiple <- mice(data_missing)
```

```
##
## iter imp variable
## 1 1 m_Expend m_Terminal
## 1 2 m_Expend m_Terminal
## 1 3 m_Expend m_Terminal
## 1 4 m_Expend m_Terminal
```

```
## 1 5 m_Expend m_Terminal
## 2 1 m_Expend m_Terminal
## 2 2 m_Expend m_Terminal
## 2 3 m_Expend m_Terminal
## 2 4 m_Expend m_Terminal
## 2 5 m_Expend m_Terminal
## 3 1 m_Expend m_Terminal
## 3 2 m_Expend m_Terminal
## 3 3 m_Expend m_Terminal
## 3 4 m_Expend m_Terminal
## 3 5 m_Expend m_Terminal
## 4 1 m_Expend m_Terminal
## 4 2 m_Expend m_Terminal
## 4 3 m_Expend m_Terminal
## 4 4 m_Expend m_Terminal
## 4 5 m_Expend m_Terminal
## 5 1 m_Expend m_Terminal
## 5 2 m_Expend m_Terminal
## 5 3 m_Expend m_Terminal
## 5 4 m_Expend m_Terminal
## 5 5 m_Expend m_Terminal
```

```
imp_multiple
```

```
## Class: mids
## Number of multiple imputations: 5
## Imputation methods:
##   Outstate  m_Expend m_Terminal
##   ""        "pmm"      "pmm"
## PredictorMatrix:
##           Outstate m_Expend m_Terminal
## Outstate      0      1      1
## m_Expend      1      0      1
## m_Terminal    1      1      0
```

```
# fits regression separately on all 5 imputed datasets
```

```
lm_multiple <- with(imp_multiple, lm(m_Expend ~ Outstate))
```

```
# The `lm_multiple` object contains the regression summaries for each of the 5
# datasets. The new object `lm_multiple` is actually of class `mira`, which
# stands for Multiply Imputed Repeated Analyses.
```

```
# combines those 5 sets of results using Rubin's pooling rules
```

```
pool_multiple <- pool(lm_multiple)
```

```
summary(pool_multiple)
```

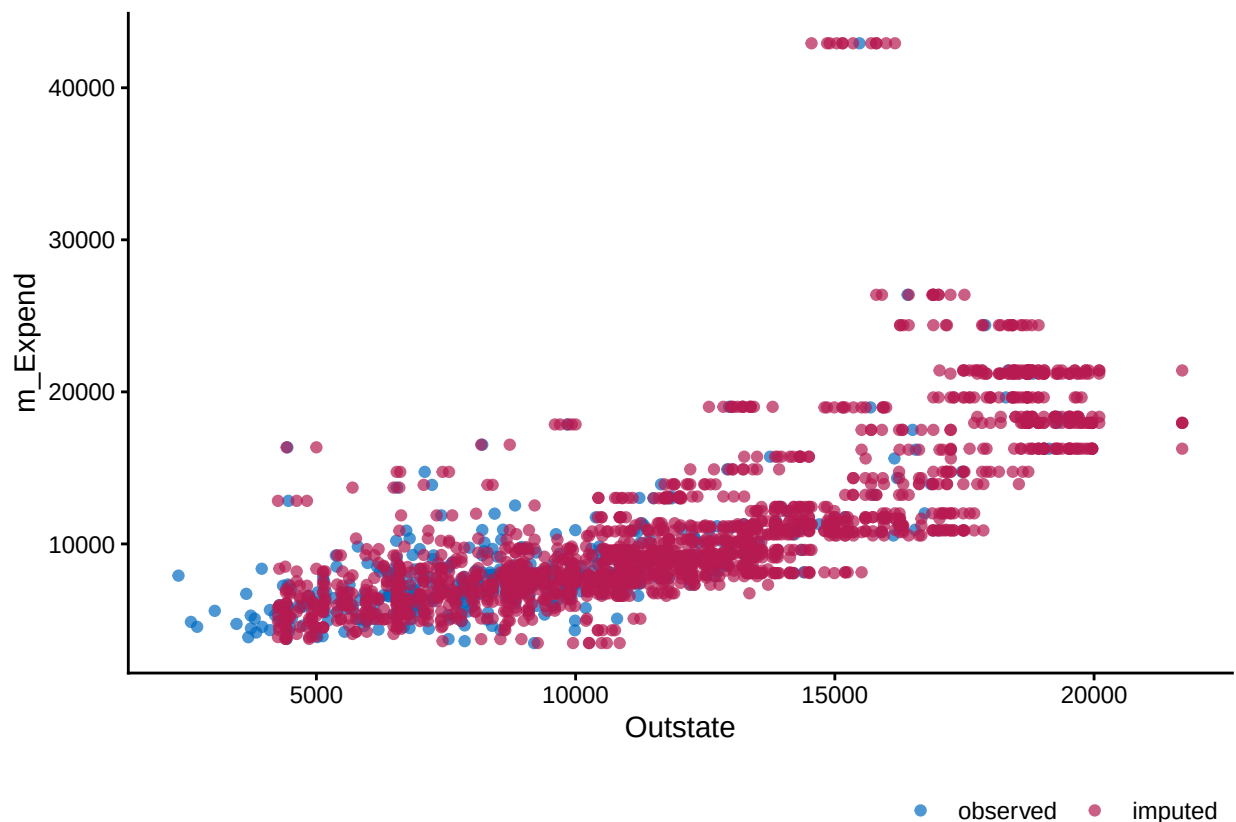
```
##           term      estimate  std.error statistic      df      p.value
## 1 (Intercept) 660.1969423 465.57223416  1.418033 14.80134 0.1768906399981168
## 2   Outstate   0.8532637   0.04434059 19.243399 11.95165 0.0000000002327278
```

```
# These results show the regression coefficient and corresponding standard
# error obtained after pooling the estimates from the five regression models
# applied on the five imputed datasets.

# Coefficient (0.853), very close to the true coefficient (0.873) by far the
# closest of any method
# Standard error (0.0443), actually larger than the true standard error (0.0345),
# which is the expected and correct behavior. It honestly reflects the added
# uncertainty from having real missing data, rather than artificially shrinking
# it like the ad hoc methods did.

# The complete pooled object gives the relevant pooled regression coefficients
# and parameters, as well as the fraction of information about the coefficients
# missing due to nonresponse (`fmi`) and the proportion of the variation
# attributable to the missing data (`lambda`). The pooled fit object is of
# class `mipo`, which stands for multiply imputed pooled object.*
```

```
imp_multiple |>
  ggmlc(aes(x = Outstate, y = m_Expend)) +
  geom_point()
```



```
# imputed points blend naturally with observed points across the whole range
# Here we take all uncertainty with respect to the imputations into account
# (i.e. uncertainty with respect to the values and with respect to the
# coefficients).
```

```
# Moreover, we impute multiple times, so that we can assess the variability
# between different data sets (on the level of the estimated model parameters).
# This is often the best approach when interest is in making inferences with
# respect to the imputed data.
```

Comparisson

```
data.frame("method" = c("complete data",
                        "listwise deletion",
                        "regression imputation",
                        "stochastic regression imputation",
                        "multiple imputation"),
           "coefficient" = c(summary(lm_complete)$coefficients[2,1],
                             summary(lm_listwise)$coefficients[2,1],
                             summary(lm_regression)$estimate[2],
                             summary(lm_stochastic)$estimate[2],
                             summary(pool_multiple)$estimate[2]),
           "standard error" = c(summary(lm_complete)$coefficients[2,2],
                                summary(lm_listwise)$coefficients[2,2],
                                summary(lm_regression)$std.error[2],
                                summary(lm_stochastic)$std.error[2],
                                summary(pool_multiple)$std.error[2]))
```

```
##               method coefficient standard.error
## 1      complete data    0.8732488      0.03449491
## 2    listwise deletion    0.7021111      0.04153479
## 3    regression imputation    0.7034919      0.01782889
## 4 stochastic regression imputation    0.7148659      0.02565823
## 5      multiple imputation    0.8532637      0.04434059
```

```
# When comparing the results from the different approaches, it can be
# concluded that in case of a dataset where the missingness follows the
# mechanism Seen Data Dependent, the estimate obtained after applying multiple
# imputation is closest to the original estimate obtained from the complete
# dataset.
#In addition, it can be concluded that the standard error obtained using multiple
# imputation is somewhat larger than the standard error obtain from the complete
# dataset. This is due to the fact that when the standard errors are pooled over
# the different imputations, not only the average is
# taken, but the variance over the differences is also incorporated i.e. the
# standard error also incorporates uncertainty due to the missing values.
# In contrast, the results obtained using listwise deletion, regression imputation
# and stochastic imputation all show a biased estimate, while regression imputation
# and stochastic imputation also show a standard error that is too low.
```

^